

TimeArch Greenfield Architecture Lifecycle

Process figures (KERKIS modeling)

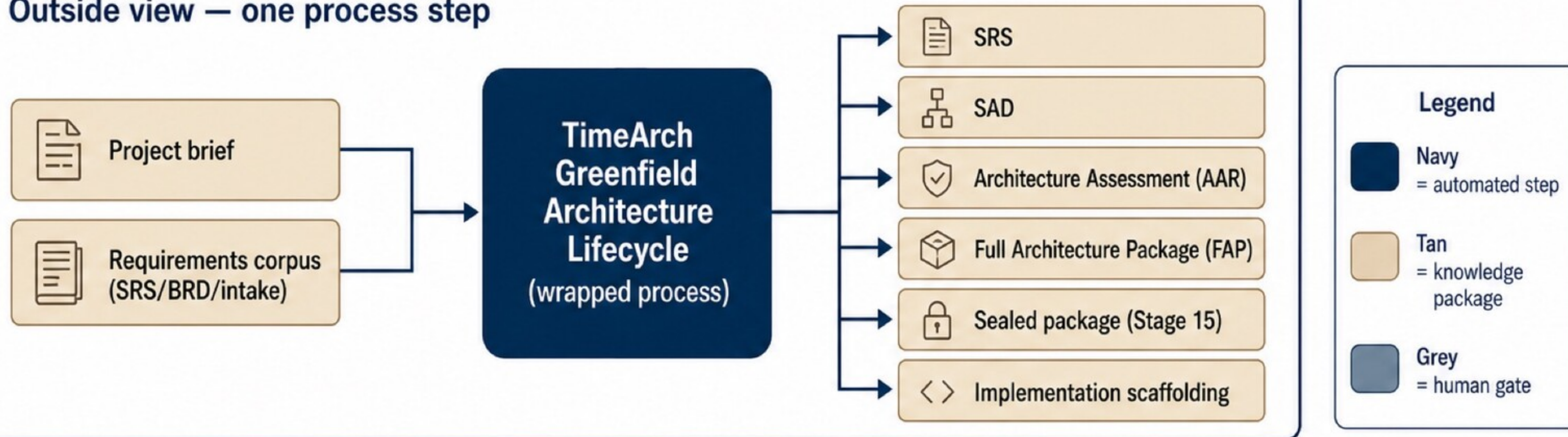
Requirements Architecture Validation Delivery

ResearchAnonymized/TimeArch · docs/modeling

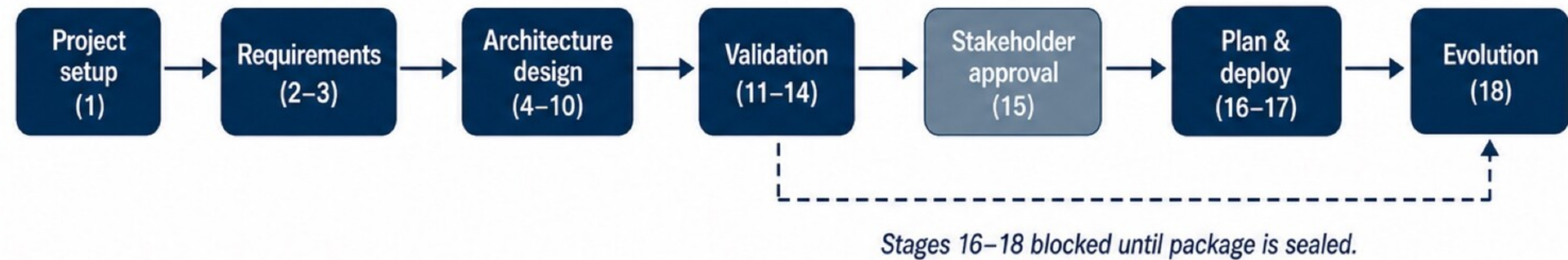
TimeArch Greenfield Architecture Lifecycle — one wrapped process

What neighbors see vs what happens inside

Outside view — one process step

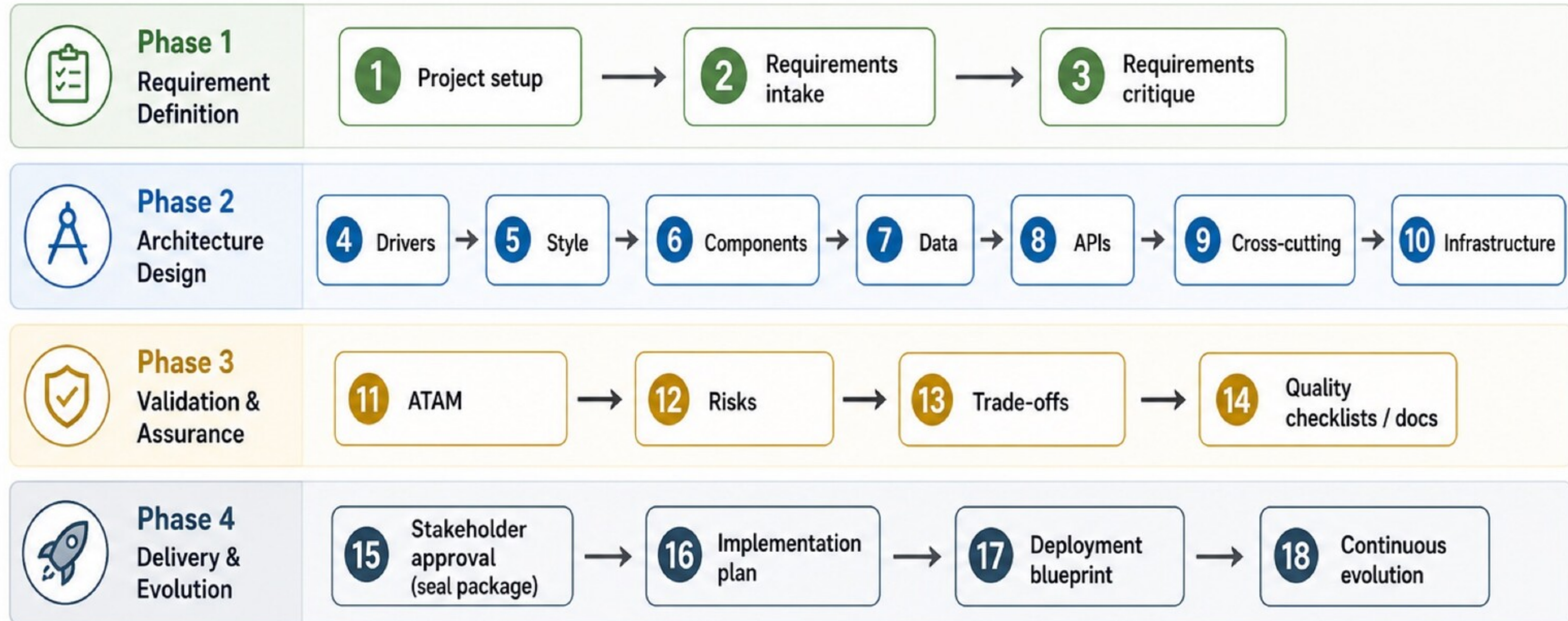


Inside view — what happens inside



Greenfield path — step by step

18 stages in four phases



Knowledge packages flow stage to stage. Stage 15 seal unlocks coding stages.



Eligibility

- Business need approved
- Initial stakeholders identified
- Governance and budget in place



Ordering

- Do stages in sequence
- Complete all steps in a phase
- Do not skip ahead



Done when

- Stage 15 seal is issued
- All packages are complete
- Ready to start coding

Who does the work in Greenfield

Code, AI, and humans across the 18-stage lifecycle



FN Code worker



Persist & lock stages



Assemble SRS/SAD/AAR/FAP documents



Package-lock gate



Deterministic checks

Examples

- Persist Stage 2 Requirement Analysis
- Lock Stage 2 with hash & timestamp
- Assemble SRS from Stage 2 package
- Assemble SAD from Stage 3 package
- Assemble AAR from Stage 7 package
- Assemble FAP from Stage 15 package
- Run deterministic validation checks
- Create package-lock gate artifact



AG AI worker



Requirement Analysis



Architecture design agents
(style, components, data, APIs...)



ATAM/risk agents



Code generation & validation



Nested Planner → Executor → Critic
on stages 2, 3, 7

Examples

- Analyze requirements (Stage 2)
- Design architecture (Stage 3)
- Run ATAM & risk analysis (Stage 7)
- Generate code skeletons & configs
- Validate outputs and produce findings
- Refine via Planner → Executor → Critic on Stages 2, 3, 7



HU Human



Project author
(setup & intake)



Architect
(run agents, accept/dismiss findings, lock stages)



Stakeholder approver
(Stage 15 seal)

Examples

- Define project goals & intake inputs
- Configure agents and stage scope
- Review AI findings and evidence
- Accept or dismiss findings
- Lock stages to create immutable record
- Approve final package at Stage 15

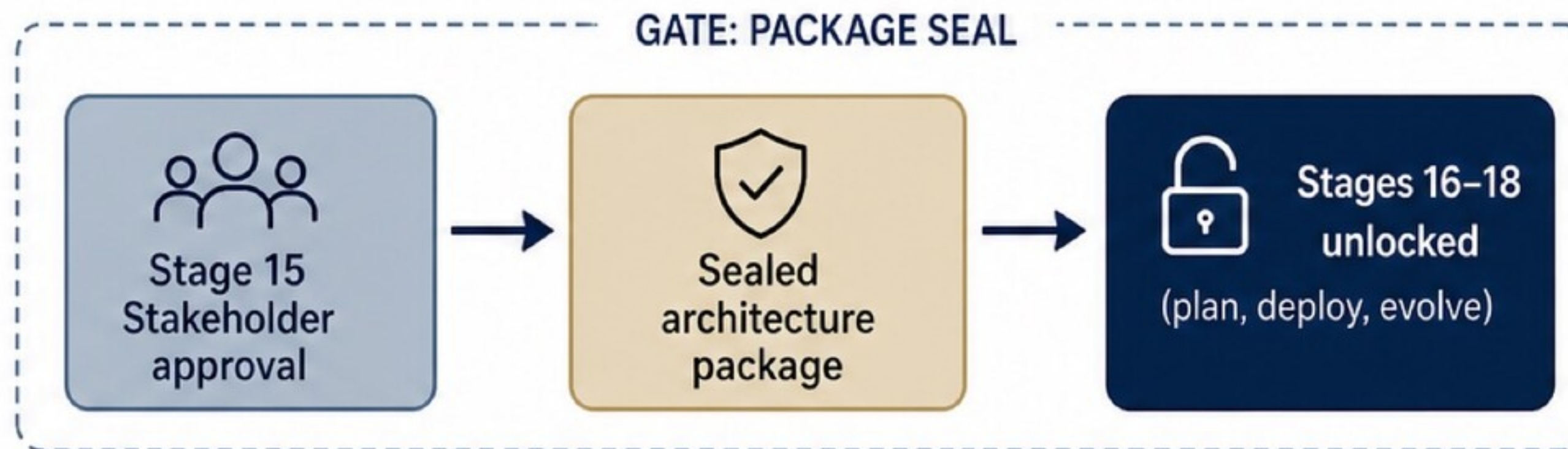
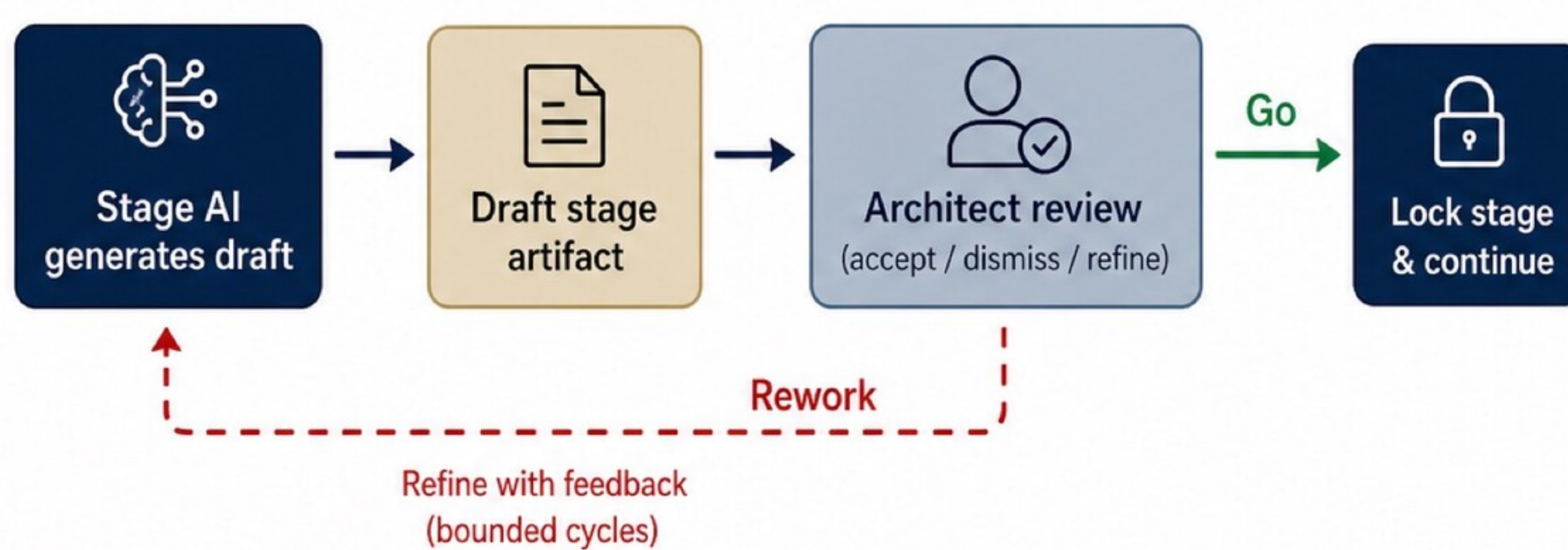


Substitutable workers; knowledge lives in packages not in the worker.



Human review and package seal

How Greenfield gates quality before code



 **Note: Without seal, implementation agents return blocked.**

-  **1. Stage locks**
 - Stages lock only after human review and approval.
-  **2. Critique (HITL)**
 - Architects accept, dismiss, or refine AI drafts.
-  **3. Challenger triage (optional)**
 - Optional challenger agents raise concerns for review.
-  **4. Package seal**
 - Stage 15 approval produces a sealed architecture package.
-  **5. Implementation gate**
 - Only sealed packages unlock Stages 16–18 for execution.

One AI step in detail — Analyze requirements

Stage 2 · Requirement Analysis Agent

Analyze requirements (AI step)



Role

Requirement
Analysis Agent



Task

Structure gaps, risks,
recommendations
from the corpus



Rules

Cite only given
requirements;
follow stage prompt



References

Project brief,
requirements rows,
knowledge snippets



Requirements corpus

All given requirements
(rows), project brief,
and knowledge snippets.

Prompt assembly



System ← Role



User ← Task + Rules +
References + corpus



Requirements analysis artifact

Includes findings (gaps,
risks), recommendations,
and traceable citations.

Parents: Requirements
corpus; Stage prompt



On stages 2/3/7 may run **Planner** → **Executor** → **Critic**
inside this step to improve quality.



Declares I/O

Consumes requirements
corpus, brief, and
knowledge snippets.
Produces requirements
analysis artifact.



Fires when inputs ready

Executes when all
required inputs are
available.



Records parents

Output artifact records
its parent inputs for
traceability.



Bounded refine cycles

Performs a limited
number of refine cycles
per governance
settings.

Nested analysis inside Greenfield

Agentic stage delivery and challenge loop

